EXAM DOMAIN AND WEIGHTINGS

The following table contains the domains and weightings covered on the exam. It is not a comprehensive listing of all the content that will be presented on the exam.

~71%

Domain	Domain Weightings
1.0 Snowflake AI Data Cloud Features and Architecture	31%
2.0 Account Management and Data Governance	20%
3.0 Data Loading, Unloading, and Connectivity	18%
4.0 Performance Optimization, Querying, and Transformation	21%
5.0 Data Collaboration	10%

1.1 Describe and use the Snowflake architecture

- Cloud Services layer
- Compute layer
- Database Storage layer

4 Ques

- Compare and contrast the different Snowflake editions

2 Ques

1.2 Use Snowflake Interfaces and tools

- Snowsight
- Snowflake CLI

2 Ques

- IDE integrations (e.g., Visual Studio Code)

1.3 Differentiate Snowflake object hierarchy and types

- Organization and account objects

1 Ques

- Database objects
  - Stages
  - Schemas
  - Tables
  - Views
  - User-Defined Functions (UDFs)
  - File formats
  - Stored procedures
  - Pipes
  - Shares
  - Sequences
  - ML models
  - Applications

4 Ques

- Session and context variables
  - Parameter hierarchy
  - Parameter precedence

1 Ques

1.4 Configure virtual warehouses

- Types
  - Snowpark Optimized
  - Standard (Gen 1 and Gen 2)
  - Default warehouse for Notebooks* (Feature will not be tested until it's globally GA)

2 Ques

- Scaling policies

2 Ques

- Warehouse type and configurations based on use cases
  - Ad-hoc queries
  - Data loading
  - BI and reporting
- Best practices
  - Sizing (up, down)
  - Scaling (in, out)
  - Auto-Suspend
  - Workloads
    - Different teams
    - High concurrency
    - Complex queries

3 Ques

1.5 Explain Snowflake storage concepts

- Micro-partitions

1 Ques

- Data clustering

1 Ques

- Table types
  - Permanent
  - Temporary
  - Transient
  - Apache Iceberg™
  - External
  - Dynamic
- Views types
  - Standard
  - Materialized
  - Secure

4 Ques

1.6 Explain AI/ML and application development features

- Snowflake Notebooks
- Streamlit in Snowflake
- Snowpark

2 Ques

- Snowflake Cortex
  - AI SQL functions
  - Cortex Search
  - Cortex Analyst
  - Document AI

1 Ques

- Snowflake ML

2.1 Explain Snowflake security model and principles

- Role-Based Access Control (RBAC)
- Securable object hierarchy
- Discretionary access control (DAC)
- Network Policies

2Ques

- Authentication
  - Multi-Factor Authentication (MFA)
  - Federated Authentication
  - Single Sign-on (SSO)
  - OAuth
  - Key-pair authentication

1 Ques

- System-defined roles
- Functional roles
  - Account roles
  - Database roles
  - Custom roles
- Secondary roles
- Account identifiers
- Logging and tracing

3 Ques

2.2 Define data governance features and how they are used

- Data masking
  - Row-level security
  - Column-level security

2 Ques

- Object tagging

2 Ques

- Privacy policies
- Trust Center
- Encryption key management
- Alerts
- Notifications
- Data replication and failover
- Data lineage

3 Ques

2.3 Explain monitoring and cost management

5 Ques

- Resource Monitors
  - Cost and warehouse monitoring
- Calculating virtual warehouse credit usage
- ACCOUNT_USAGE schema

3.1 Perform data loading and unloading

- File formats 1 Ques
- Create and use stages
  - Internal stages
  - External stages
  - Server-side encryption
  - Directory tables

  4 Ques
- COPY INTO command 3 Ques
- Error handling options

3.2 Perform automated data ingestion

- Snowpipe
- Snowpipe streaming

  2 Ques
- Streams
- Tasks

  1 Ques
- Dynamic tables 1 Ques
- Openflow (Feature will not be tested until it's globally GA)

3.3 Identify the different Snowflake Connectors and integrations

- Snowflake drivers
- Snowflake connectors
- Storage integration
- API integration
- Git integration

  2 Ques

4.1 Evaluate query performance

- Query Performance Tuning
  - Query Profile/Query insights
    - Bytes spilled to storage
    - Inefficient pruning
    - Exploding joins
    - Queuing

3 Ques

- SNOWFLAKE.ACCOUNT_USAGE views (Snowflake database views)
  - Query attribution
  - Query history

2 Ques

- Workload management best practices
  - Grouping similar workloads

1 Ques

4.2 Optimize query performance

- Query acceleration service 3 Ques
- Search optimization service 2 Ques
- Clustering keys
- Materialized views

2 Ques

4.3 Use Snowflake caching

- Query result cache
- Metadata cache
- Warehouse cache

4 Ques

4.4 Perform data transformation techniques

- Using data
  - Structured
  - Semi-structured
  - Unstructured

2 Ques

- Aggregate functions 1 Ques
- Applying SQL for query optimization
- Window functions

5.1 Explain data collaboration and protection

- Data replication and failover
- Secure data sharing features

1 Ques

- Cloning
- Time Travel
- Fail-safe

3 Ques

5.2 Explain Snowflake's data sharing capabilities

- Accounts
  - Provider
  - Consumer
  - Reader accounts

2 Ques

- Secure Data Sharing
- Sharing and resharing
- Direct shares

2 Ques

- Data clean rooms

5.3 Share data using the Snowflake Marketplace and listings

1 Ques

- Snowflake Marketplace
- Listings
  - Private
  - Public

2 Ques

- Native Apps

The 'Negative' Rule

Exam Question: A data architect is looking to scale a virtual warehouse to handle a massive, complex JOIN operation involving billions of rows. Which of the following statements is **TRUE** regarding Snowflake's concurrency and scaling?

A. Snowflake requires a manual restart of the warehouse to change the size from Small to X-Large. ✗

B. Multi-cluster warehouses can automatically scale out to handle increased concurrency without manual intervention.

C. Snowflake's performance significantly degrades when more than 10 users query the same table simultaneously. ✗

D. Data is physically moved between storage nodes every time a warehouse is resized, causing a temporary lock on the data.

The Relevance Filter

Sample Example: An administrator needs to ensure that a specific set of sensitive tables can only be queried by users in the `HR_FULL_ACCESS` role, regardless of which warehouse they are using. Which Snowflake feature should be used to satisfy this requirement?

- **A.** Multi-cluster Warehouses ✗
- **B.** Object-level Access Control (RBAC)
- **C.** Search Optimization Service ✗
- **D.** Resource Monitors ✗

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  - Cortex Analyst
  - Document AI
- Snowflake ML

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